

A disproof of Conjecture 00000003462

K_6 breaks the bound $(5v - 2)/3$, and no six-doubly-critical graph attains it

1 The conjecture

Conjecture 00000003462 reads (English version):

Definition: Doubly critical graphs: k -coloring edge-critical graphs where deleting any two vertices lowers the chromatic number, with Gallai's bound the edge upper bound of critical graphs. *Conjecture:* The Gallai upper bound for six-doubly-critical graphs improves to $(5v - 2)/3$, with the improved bound uniquely attained by Kostochka–Yancey extremal graphs; the classification of doubly critical and four-critical merges into one extremal family at girth at least five.

The Chinese version defines the class differently: “shuang linjie tu: k -ranse bian linjie tu manzu renyi liang dian de shanchu shi seshu jiang yi”, i.e. k -chromatic edge-critical graphs in which deleting any two vertices makes the chromatic number *drop by one*. Its conjecture is the same: the bound improves to $(5v - 2)/3$, and the improved bound is attained uniquely by Kostochka–Yancey type extremal graphs (“you Kostochka–Yancey xing ji tu wei yi qude”).

Structure of the claim. With D the class of six-doubly-critical graphs, the conjecture is the conjunction of (B) every $G \in D$ with v vertices has $e \leq (5v - 2)/3$ edges; (A) the bound is attained by some $G \in D$; (U) it is attained only by Kostochka–Yancey extremal graphs; (C) a girth-five classification clause. We refute $(B) \wedge (A)$ under every reading below, so the conjunction fails whatever (U) and (C) mean.

Readings of the class ($k = 6$). All graphs are finite and simple, $G - x - y$ is the induced subgraph on $V \setminus \{x, y\}$, and “edge-critical” means that deleting any edge lowers χ .

- (EN) English: $\chi(G) = 6$, G is edge-critical, and $\chi(G - x - y) < \chi(G)$ for all distinct x, y .
- (ZH) Chinese, “drops by one” read as *exactly* one: the same with $\chi(G - x - y) = \chi(G) - 1$. (“Drops by at least one” is (EN).)
- (EL) The standard notion (Erdos–Lovasz): G is connected, $\chi(G) = 6$, G is edge-critical, and $\chi(G - x - y) = \chi(G) - 2$ for every *edge* xy .

Since $\chi(G - x - y) = 5 < 6$ under (ZH), the class (ZH) is contained in (EN). Gallai's bound is a lower bound in the literature, so we also treat the variant (B'): every $G \in D$ has $e \geq (5v - 2)/3$.

Results. (1) K_6 is in (EN) and (EL) and has $15 > 28/3$ edges: (B) is false. (2) Every graph in (EN), hence in (ZH), has $e > (5v - 2)/3 + 1$: no graph in either class attains $(5v - 2)/3$, even up to rounding, so (A) is false. Thus $(B) \wedge (A)$ and $(B') \wedge (A)$ fail under (EN) and (ZH), and (B) fails under (EL).

2 Definitions and sources

Wikipedia (https://en.wikipedia.org/wiki/Critical_graph): “a critical graph is an undirected graph all of whose proper subgraphs have smaller chromatic number.” Every critical graph is edge-critical, so all our statements about edge-critical graphs apply to critical ones. The same page: “A double-critical graph is a connected graph in which the deletion of any pair of adjacent vertices decreases the chromatic number by two.” Kostochka and Yancey (<https://arxiv.org/abs/1209.1050v1>) study “the minimum number of edges in an n -vertex k -critical graph” and write that “The result improves the classical bounds by Gallai and Dirac”, so Gallai’s bound is a lower bound on e .

3 Proof

Lemma 1. K_6 has $\chi = 6$; deleting any edge ab gives a 5-colourable graph; deleting any two vertices leaves K_4 with $\chi = 4$. K_6 is connected. Hence K_6 belongs to (EN) and (EL) (not to (ZH)), and $e(K_6) = 15 > 28/3 = (5 \cdot 6 - 2)/3$.

Proof. $\chi(K_n) = n$. For $K_6 - ab$, colour every vertex $w \neq b$ by itself and b by a ; the colour set $\{w : w \neq b\}$ has 5 elements, and the only pair that gets equal colours is $\{a, b\}$, which is no longer an edge. $\square$

Lemma 2. If $\chi(G) = 6$ and G is edge-critical, every vertex v with a neighbour u has $\deg v \geq 5$.

Proof. $G - uv$ has a proper 5-colouring C . If $\deg v \leq 4$, some colour c is not used by C on the neighbours of v . Recolour v with c . Edges at v are now proper, and every other edge of G is an edge of $G - uv$, so this is a proper 5-colouring of G , contradicting $\chi(G) = 6$. $\square$

Lemma 3. If G is in (EN), G has at most one isolated vertex.

Proof. G has an edge zz' (otherwise G is 1-colourable). If $x \neq y$ were both isolated, then $z \notin \{x, y\}$, and $\chi(G - x - y) < 6$ gives a 5-colouring of $G - x - y$. Give x and y the colour of z . Since x, y have no edges, this is a proper 5-colouring of G , a contradiction. $\square$

Theorem 4. Every G in (EN), and hence every G in (ZH), satisfies $5v + 1 < 3e$, i.e. $e > (5v - 2)/3 + 1$.

Proof. By Lemmas 2 and 3, $\sum_v (\deg v + 5[\deg v = 0]) \geq 5v$, so $2e = \sum_v \deg v \geq 5v - 5$. Also $v \geq \chi(G) = 6$. Then $6e \geq 15v - 15 > 10v + 2$ because $5v > 17$. $\square$

Corollary 5. (B) fails for (EN) and (EL), by K_6 . Under (EN) and (ZH), no graph attains $(5v - 2)/3$: not exactly, and not within distance < 1 (so neither $\lfloor (5v - 2)/3 \rfloor$ nor $\lceil (5v - 2)/3 \rceil$). Hence (A) fails, and the conjecture $(B) \wedge (A) \wedge (U) \wedge (C)$ and its lower-bound variant $(B') \wedge (A) \wedge (U) \wedge (C)$ are false under (EN) and (ZH), for any (U) and (C). Under the lower-bound variant the inequality (B') itself holds, strictly.

4 Lean 4 formalization

The directory `lean/` is a Lean 4 project (Lean v4.33.1, Mathlib v4.33.1). `Conjecture3462.lean` imports `Conjecture3462/Basic.lean` (331 lines, namespace `C3462`). Graphs are Mathlib `SimpleGraphs`; chromatic numbers are Mathlib’s `chromaticNumber` (values in $\mathbb{N}_\infty$); $G - x - y$ is `G.induce ({x, y}ᶜ : Set V)`; edge deletion is `G.deleteEdges {e}`.

Definitions.

- `IsEdgeCritical G`: for all $e \in G.\text{edgeSet}$, $(G.\text{deleteEdges } \{e\}).\text{chromaticNumber} < G.\text{chromaticNumber}$.
- `IsDoublyCriticalEN k G`: $G.\text{chromaticNumber} = k$, `IsEdgeCritical G`, and for all $x \neq y$, $(G.\text{induce } \{x, y\}^c).\text{chromaticNumber} < G.\text{chromaticNumber}$.
- `IsDoublyCriticalZH k G`: the same with $(G.\text{induce } \{x, y\}^c).\text{chromaticNumber} + 1 = G.\text{chromaticNumber}$.
- `IsDoublyCriticalEL k G`: $G.\text{Connected}$, $G.\text{chromaticNumber} = k$, `IsEdgeCritical G`, and for all $x \neq y$ with $G.\text{Adj } x \ y$, $(G.\text{induce } \{x, y\}^c).\text{chromaticNumber} + 2 = G.\text{chromaticNumber}$.
- `GraphClass := (n : ℕ) → SimpleGraph (Fin n) → Prop` (every finite graph is isomorphic to one on $\text{Fin } n$); `EN6`, `ZH6`, `EL6` are the three classes with $k = 6$.
- `ImprovedBound D`: every G in D on $\text{Fin } n$ has $(G.\text{edgeSet}.\text{ncard} : \mathbb{R}) \leq (5 * n - 2) / 3$. `ImprovedLowerBound D`: $\geq$.
- `BoundAttained D`: some G in D has $\text{ncard} = (5 * n - 2) / 3$; `BoundAttainedUpToRounding D`: some G has $|\text{ncard} - (5 * n - 2) / 3| < 1$.
- `Conjecture D KY girthClause := ImprovedBound D ∧ BoundAttained D ∧ (every G in D attaining the bound satisfies KY n G) ∧ girthClause`; `ConjectureLower`: the same with `ImprovedLowerBound`. `KY` and `girthClause` are arbitrary.

Theorems.

- `top_chromaticNumber`, `top_edgeCritical`, `top_deleteTwo`, `top_EN`, `top_EL` (connectedness from Mathlib's `connected_top`), `top_edges`: Lemma 1 for $(\top : \text{SimpleGraph } (\text{Fin } 6))$.
- `not_improvedBound_of_top` (any class containing K_6), `not_improvedBound_EN`, `not_improvedBound_EL`: (B) fails.
- `ZH_to_EN`: $(\text{ZH}) \subseteq (\text{EN})$.
- `degree_ge_five`: Lemma 2. `not_adj_of_degree_eq_zero`, `mem_compl_pair`, `colorable_of_induce`: the extension step of Lemma 3.
- `edges_gt : 5 * Fintype.card V + 1 < 3 * G.edgeSet.ncard` for `IsDoublyCriticalEN 6 G` (Theorem 4); `edges_gt_real`.
- `SubEN D`: every graph of D satisfies `IsDoublyCriticalEN 6`; `subEN_EN`, `subEN_ZH`. For every D with `SubEN D`: `not_attainedUpToRounding`, `improvedLowerBound` and `not_conjecture` ($\neg \text{Conjecture } D \text{ KY } \text{girthClause} \wedge \neg \text{ConjectureLower } D \text{ KY } \text{girthClause}$); `attained_of_exact`: Corollary 5.
- Main theorem `conjecture_00000003462_false`, the conjunction of
 - `IsDoublyCriticalEN 6  $\top$` , `IsDoublyCriticalEL 6  $\top$` , $\top.\text{edgeSet}.\text{ncard} = 15$ (on $\text{Fin } 6$);

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- ¬ImprovedBound EN6, ¬ImprovedBound EL6;
- ¬BoundAttainedUpToRounding EN6, ¬BoundAttainedUpToRounding ZH6;
- for all KY girthClause: ¬Conjecture EN6 KY girthClause, ¬Conjecture ZH6 KY
  girthClause, ¬ConjectureLower EN6 KY girthClause, ¬ConjectureLower ZH6 KY
  girthClause.

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Axiom audit. There is no sorry, no native_decide and no added axiom. #print axioms gives

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'C3462.conjecture_00000003462_false' depends on axioms:
  [propext, Classical.choice, Quot.sound]

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To check: cd lean; lake exe cache get; lake build.